

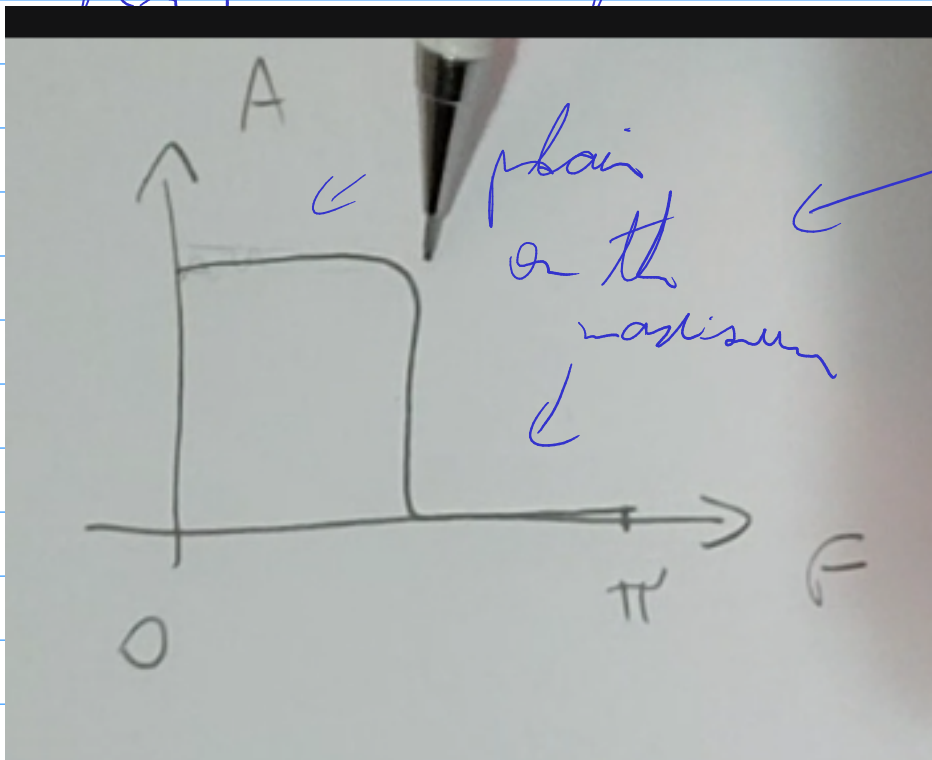
08
05
26

Darb-4 $\rightarrow$ Darbules filter with support size 4
 $\uparrow$
some arbitrary part of the number of null moments

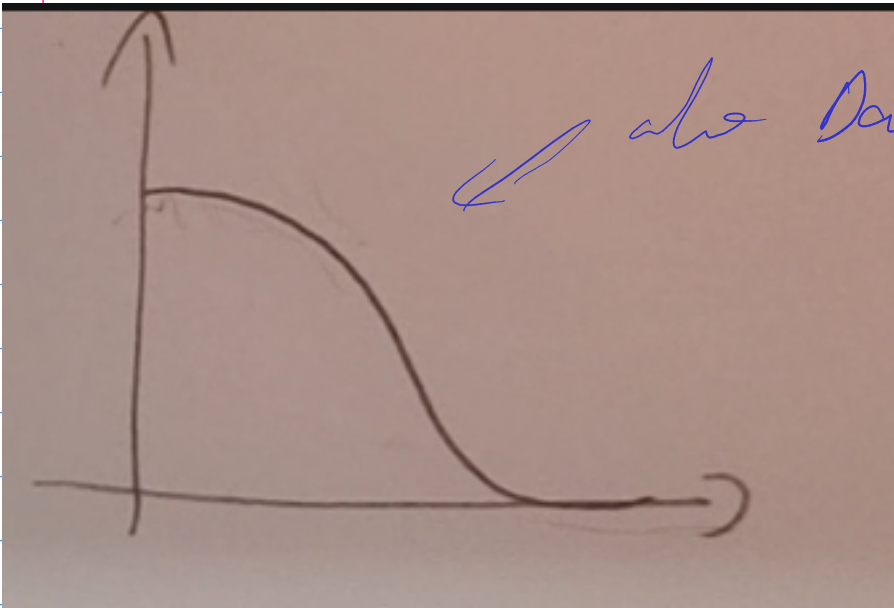
Darb-4 $\rightarrow$ has 2 null moments to get a half band cut

Symmetris $\rightarrow$ a family of filter

the response is frequency $(|F(x)|)$
(the way it returns) is one of the characteristics of a filter

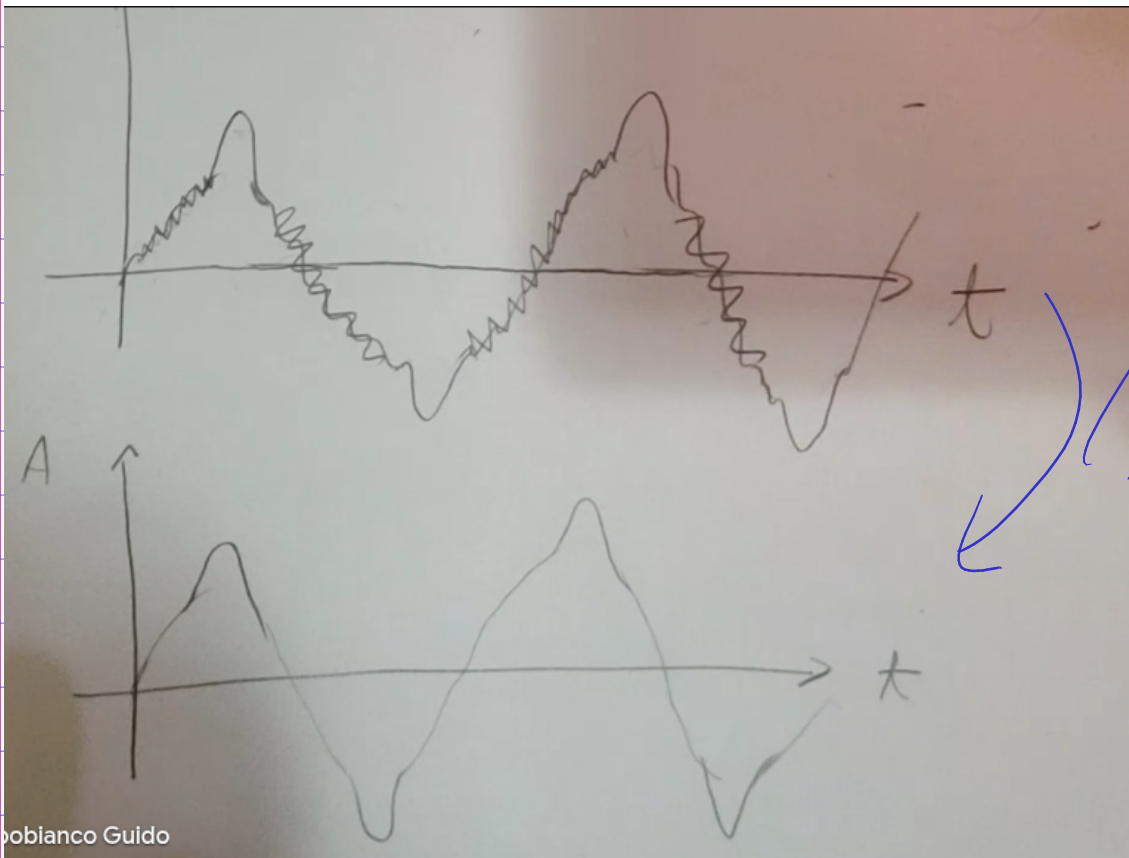


Darb response is frequency



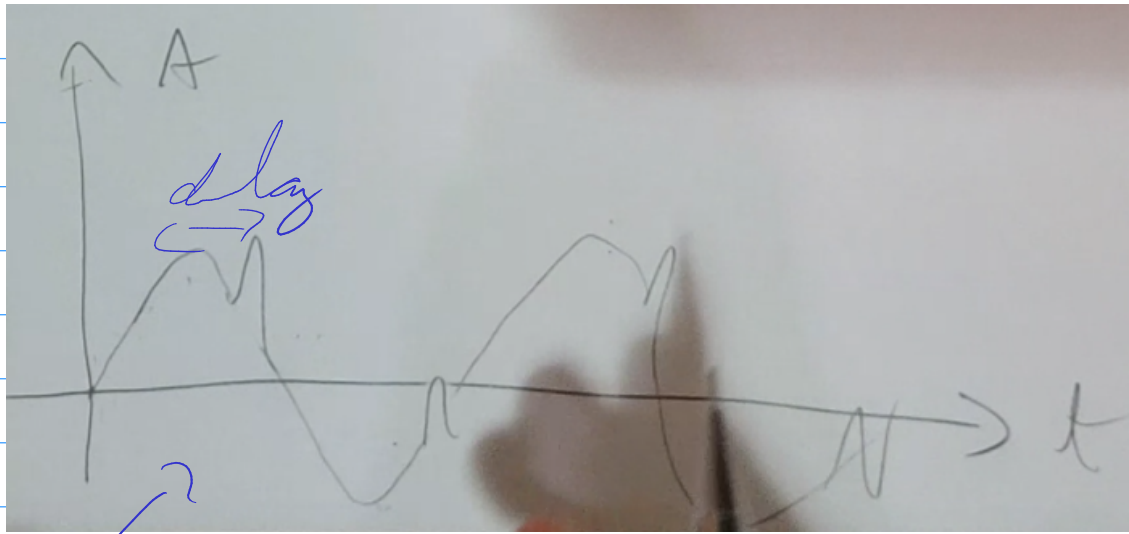
also Danch

The plan in this can represent
the delay it's gain to frequency in
the frequency after filtering



Filtered
signal

Then fluctuation we have removed
 present a higher behavior in this
 position will be the same as before
 with gaps



Filtering is only a removal of
 signal which can also have fluctua-
 tion. Then higher peaks, or after
 then filtering, depending on the filter
 response in plan can shift. Then
 fluctuation we are left with
 with the piece behavior, what that
 high delay

Pauli - is not linear in plan

Symmetry $\rightarrow$ are planes in the
 $\hookrightarrow$ maximum support is multiple of
 $\hookrightarrow$ 8 in general
 $\hookrightarrow$ short series is
 plane expansion
 $\hookrightarrow$ are orthogonal
 $\hookrightarrow$ Vanishing moment centered

--- wavelet families: Symmlets ---

► **Numerical Example:** for $N = 8$, we have the following nonlinear system of $N = 8$ equations in $N = 8$ unknowns:

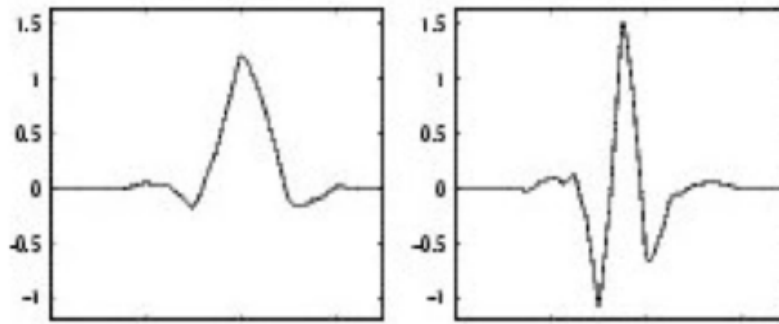
$\wedge$ explicit
 N equations

$$\left\{ \begin{array}{ll} (g_0)^2 + (g_1)^2 + (g_3)^2 + \dots + (g_7)^2 & = 1 \quad (\text{filter energy}) \\ g_0 + g_1 + g_2 + g_3 + \dots + g_7 & = 0 \quad (\text{1st vanishing moment}) \\ -3g_0 - 2g_1 - 1g_2 + 0g_3 + \dots + 4g_7 & = 0 \quad (\text{2nd vanishing moment}) \\ 9g_0 + 4g_1 + 1g_2 + 0g_3 + \dots + 16g_7 & = 0 \quad (\text{3rd vanishing moment}) \\ -27g_0 - 8g_1 - 1g_2 + 0g_3 + \dots + 64g_7 & = 0 \quad (\text{4th vanishing moment}) \\ g_0g_2 + g_1g_3 + g_2g_4 + g_3g_5 + \dots + g_5g_7 & = 0 \quad (\text{to force orthogonality}) \\ g_0g_4 + g_1g_5 + g_2g_6 + g_3g_7 & = 0 \quad (\text{to force orthogonality}) \\ g_0g_6 + g_1g_7 & = 0 \quad (\text{to force orthogonality}) \end{array} \right.$$

Its solution corresponds to the Symm filter $g[\cdot]$ with support-size $N = 8$, i.e., $g[\cdot] = \{-0.075, 0.029, 0.497, -0.803, 0.297, 0.099, -0.012, -0.032\}$. Consequently, $h[\cdot] = \{0.032, -0.012, -0.099, 0.297, 0.803, 0.497, -0.029, -0.075\}$. Although we do not need them to perform DTWTs, the scaling and wavelet functions can also be calculated, as in the other families, explicitly or by iteration.

filter value precision roughly is 10^{-4}
 (double)

--- Wavelet Families: Symmlets ---



Just as an example: [left] Symmlet scaling function for $N = 8$ and [right] Symmlet wavelet function for $N = 8$. Notably, those functions are almost-symmetric. From signal processing theory, we learn that (almost) symmetric or anti-symmetric impulse responses produce (almost) linear phase responses.

- **Remark:** Symm filters are not maximally-flat as Daub are. Thus, Symm filters frequency responses might occasionally present small

have
they
look
alike

are not maximally-flat

impulse response $\rightarrow$ the coefficient
is the more symmetrical anti-symmetric
the more the phase will be like

Weights $\rightarrow$ multiplies of 6
 $\hookrightarrow$ orthogonal

--- Wavelet Families: Coiflets ---

- Coiflet filters are specified as follows:

- Force $g[\cdot]$ to have unitary energy so that the DTWT preserves that of the input signal, i.e.,

$$\sum_{k=0}^{N-1} g_k^2 = 1 \quad ;$$

- Impose $\frac{N}{3}$ vanishing moments on the wavelet function, i.e.,

$$\sum_{k=0}^{N-1} g_k k^b = 0 \quad , (b = 0, 1, \dots, \frac{N}{3} - 1) \quad ;$$

- Impose $\frac{N}{3}$ vanishing moments on the scaling function, i.e.,

$$\sum_{k=0}^{N-1} (-1)^k \left(\frac{N}{2} - k\right)^b g_k = 0 \quad , (b = 0, 1, \dots, \frac{N}{6} - 1).$$

Notably, $\frac{N}{3}$ vanishing moments on the scaling functions can be obtained based on $\frac{N}{6}$ equations only! This is a result confirmed by Gopinath and Burrus, decades ago:

- - - Wavelet Families: Coiflets - - -

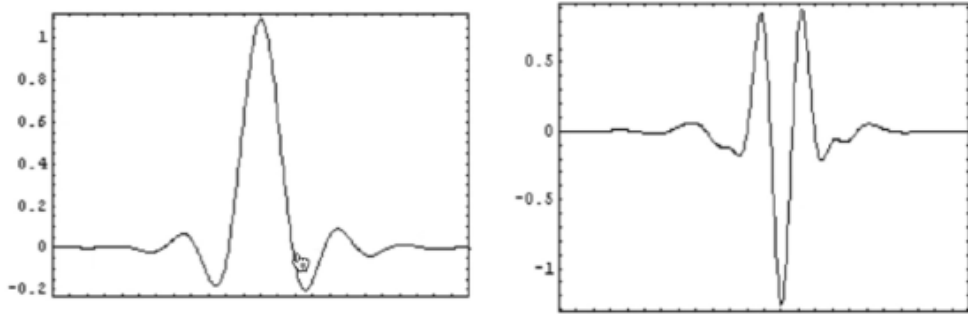
- **Numerical Example:** for $N = 6$, we have the following nonlinear system of $N = 6$ equations in $N = 6$ unknowns:

$$\begin{cases} (g_0)^2 + (g_1)^2 + \dots + (g_5)^2 &= 1 & \text{(filter energy)} \\ g_0 + g_1 + g_2 + g_3 + g_4 + g_5 &= 0 & \text{(first vanishing moment wavelet)} \\ 0g_0 + 1g_1 + 2g_2 + \dots + 5g_5 &= 0 & \text{(second vanishing moment wavelet)} \\ g_0 - g_1 + g_2 - g_3 + g_4 - g_5 &= 0 & \text{(first vanishing moment scaling)} \\ g_0g_2 + g_1g_3 + g_2g_4 + g_3g_5 &= 0 & \text{(required to force orthogonality)} \\ g_0g_4 + g_1g_5 &= 0 & \text{(required to force orthogonality)} \end{cases}$$

Its solution corresponds to the Coiflet filter $g[\cdot]$ with support-size $N = 6$, i.e., $g[\cdot] = \{\frac{-3+\sqrt{7}}{16\sqrt{2}}, \frac{-1+\sqrt{7}}{16\sqrt{2}}, \frac{14-2\sqrt{7}}{16\sqrt{2}}, \frac{-14-2\sqrt{7}}{16\sqrt{2}}, \frac{5+\sqrt{7}}{16\sqrt{2}}, \frac{-1+\sqrt{7}}{16\sqrt{2}}\}$. Consequently, $h[\cdot] = \{\frac{1-\sqrt{7}}{16\sqrt{2}}, \frac{5+\sqrt{7}}{16\sqrt{2}}, \frac{14+2\sqrt{7}}{16\sqrt{2}}, \frac{14-2\sqrt{7}}{16\sqrt{2}}, \frac{1-\sqrt{7}}{16\sqrt{2}}, \frac{-3+\sqrt{7}}{16\sqrt{2}}\}$.

- ▶ Although we do **not** need them to perform DTWTs, the scaling and wavelet functions can also be calculated, as in Haar, Daub and Symm families, explicitly or by iteration.

- - - Wavelet Families: Coiflets - - -



Just as an example: [left] Coiflet scaling function for $N = 12$ and [right] Coiflet wavelet function for $N = 12$.

- Remark: Coif filters are not maximally-flat as Daub are. Thus, Coif filters frequency responses might occasionally present small ripples in the pass- and stop-bands.

other parties: Lac, Buh, Syon, Gil

wavelet families, such as Beylkin and Vaidyanathan which are optimized to analyze audio and speech data, among others. They are not, however, so usual in practice and thus we will not study them.

non filter specific

non moment

biorthogonal family + CDF, Le Gall - Tokic
Sci. (Fig 2000)

$h[\cdot]$ and $g[\cdot]$ are not necessarily minimal, they can also have different sizes. In that case they may need to specify the $h[\cdot]$ $g[\cdot]$. This is hard, but

$$\begin{cases} (g_0)^2 + (g_1)^2 + \dots + (g_5)^2 & = 1 & \text{(filter energy)} \\ g_0 + g_1 + g_2 + g_3 + g_4 + g_5 & = 0 & \text{(first vanishing moment wavelet)} \\ 0g_0 + 1g_1 + 2g_2 + \dots + 5g_5 & = 0 & \text{(second vanishing moment wavelet)} \\ g_0 - g_1 + g_2 - g_3 + g_4 - g_5 & = 0 & \text{(first vanishing moment scaling)} \\ g_0g_2 + g_1g_3 + g_2g_4 + g_3g_5 & = 0 & \text{(required to force orthogonality)} \\ g_0g_4 + g_1g_5 & = 0 & \text{(required to force orthogonality)} \end{cases}$$

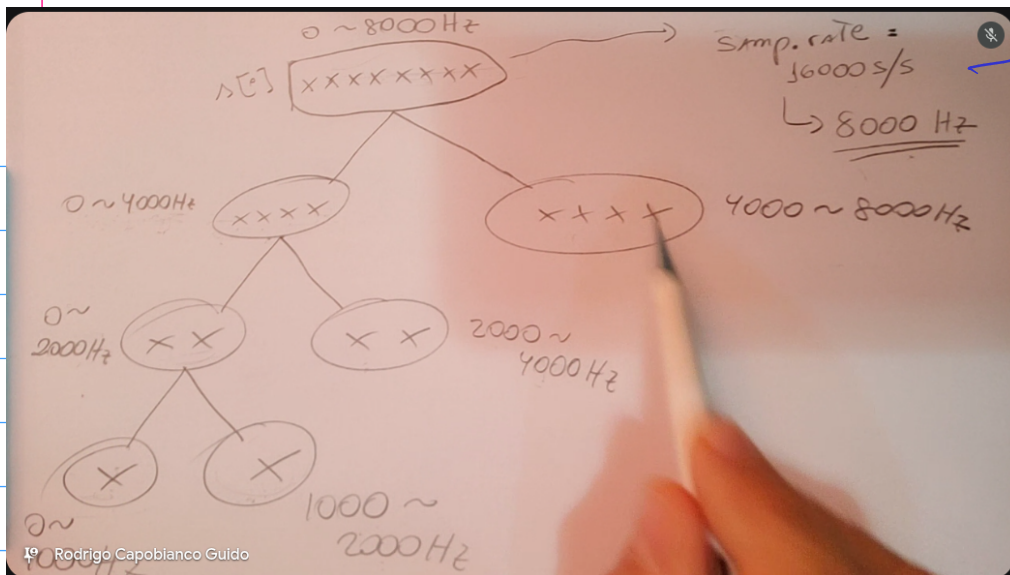
in a system with $N=6$ equations you need to ensure the orthogonality, but in that case we can get not of other and be more precise in frequency and phase responses

They also have two main functions $\hat{\phi}(x)$ and $\hat{\psi}(x)$ scaling and wavelet basis,

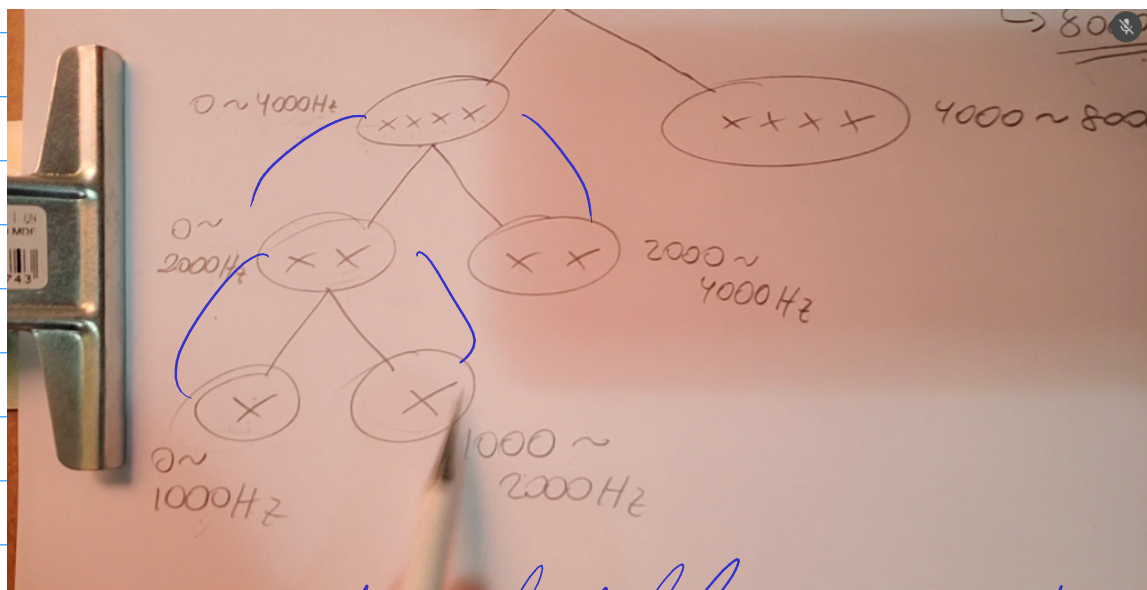
birth of $\mathcal{L}$ is given by $\mathcal{L}^{\pm}$ and $\mathcal{L}^{\pm}$ can be ordered to each other and $\mathcal{L}^{\pm}$ and $\mathcal{L}^{\pm}$ as well

time-frequency plan application 1 -

- ▶ Problem Statement: Given the pair of signals $s[\cdot]$ and $y[\cdot]$, both of size M equal to a power of two and indexed from 0 to $M - 1$, being the former and the latter the discrete-time input and its DTWT, respectively, the problem is to extract and interpret relevant information the latter contains about the former, particularly allowing for the time-support of frequencies to be accurately found.
- ▶ Solution: Differently from Fourier analysis, which converts a signal from the time to the frequency domain, the DTWT *filters* it. Thus, significant amplitudes in $y[\cdot]$ come from frequencies that originally exist in $s[\cdot]$ and were not removed by the filters. Based on such a process, the proper strategy consists of a DTWT-based filtering applied to $s[\cdot]$ followed by an inspection of the dominant amplitudes contained in the transformed signal $y[\cdot]$.



you always need the sample rate



at each level you get that in SFC might have frequencies N that half repeatedly. You may want to get precisely a frequency you have to get a range instead, because in the last node you have the range, in this example 1000 Hz , you don't know the frequency more precisely than that

Diagram illustrating the frequency spectrum of a sampled signal, showing the relationship between the sampling rate and the resulting spectrum.

Sampling rate: $f_s = 16000 \text{ s/s}$

Resulting spectrum components:

- Original signal spectrum: $0 \sim 8000 \text{ Hz}$ (indicated by a red arrow pointing to the original spectrum).
- Aliased spectrum components: $4000 \sim 8000 \text{ Hz}$ (indicated by a green arrow pointing to the aliased spectrum).
- Aliased spectrum components: $2000 \sim 4000 \text{ Hz}$ (indicated by a green arrow pointing to the aliased spectrum).
- Aliased spectrum components: $0 \sim 2000 \text{ Hz}$ (indicated by a green arrow pointing to the aliased spectrum).

Handwritten notes:

- For each sample...

he is
a balance
point
is the

The name
is 4 points,
that symmetry!

row each part
 represents a
 pair of 4 is right
 no the or
 with bigger
 amplified will
 say the part
 that the part
 has an

we must (re)state that, the more
we decompose (+ break) the more rougher
is frequently you'll have, but less the
then/positive resolution you'll have, the
uncertainty of these things

Note if we need we can keep
charging but on the left (weight
back)

If we want to have both weights,
we can find in this and then keep
charging

Weight packet $\rightarrow$ when the
discharge

A good strategy would be to test different
band filters to approximate the shape of the signal
SET.

Fluctuation energy $E \approx 0$

band maximally-flat behavior gives us a more
stable gain & frequency at the beginning before
the transition